WORKED SOLUTION

STEP II 1990, Question 15

Probability and Statistics • Pdf

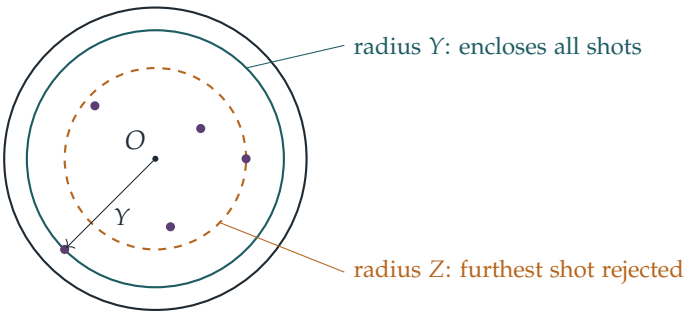
STEP II 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q15

The key idea

The smallest circle centred at O enclosing a set of shots has radius equal to the distance of the *furthest* shot, so Y is a maximum of independent distances. A maximum is at most y exactly when *every* shot is within y, which turns the cumulative distribution function into a product. After rejecting the furthest shot, the radius we want is the *second* largest distance, and its distribution function comes from counting how many shots may lie outside a given circle.



Solution

Let $R_1, \ldots, R_n$ be the distances of the n shots from O. They are independent (the shots are fired independently), and each has distribution function

$$P(R_i \leq t) = t^2, \qquad 0 \leq t \leq 1.$$

The distribution of Y

The smallest circle centred at O containing every shot has radius $Y = \max(R_1, \ldots, R_n)$. For $0 \le y \le 1$,

$$P(Y \le y) = P(R_1 \le y, \ldots, R_n \le y) = \prod_{i=1}^{n} P(R_i \le y) = (y^2)^n = y^{2n},$$

using independence for the middle step. Differentiating, the probability density function of Y is

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} y^{2n} = 2ny^{2n-1}, \qquad 0 \le y \le 1,$$

as required. (It is zero outside $[0, 1]$, and it integrates to 1, as it should.)

The area of the circle is πY^2, so

$$E(\pi Y^2) = \pi \int_0^1 y^2 \cdot 2ny^{2n-1}\,\mathrm{d}y = 2n\pi \int_0^1 y^{2n+1}\,\mathrm{d}y = \frac{2n\pi}{2n+2} = \frac{n\pi}{n+1}.$$

Rejecting the furthest shot

We need $n \ge 2$ so that some shots remain. Once the furthest shot is removed, the smallest enclosing circle centred at O has radius Z, the second largest of $R_1, \ldots, R_n$.

> **Ask yourself** How can we describe the event $Z \le z$ in terms of how many shots land outside the circle of radius z?

The second largest distance is at most z exactly when *at most one* shot lies further than z from O. Each shot independently lies within z with probability z^2, so the number of shots outside the circle of radius z is binomial, $\mathrm{B}(n, 1 - z^2)$. Hence, for $0 \le z \le 1$,

$$\begin{aligned} P(Z \le z) &= P(\text{no shot outside}) + P(\text{exactly one shot outside}) \\ &= z^{2n} + n(1 - z^2)z^{2(n-1)} \\ &= nz^{2n-2} - (n-1)z^{2n}. \end{aligned}$$

Differentiating gives the density of Z:

$$f_Z(z) = 2n(n-1)z^{2n-3} - 2n(n-1)z^{2n-1} = 2n(n-1)z^{2n-3}(1 - z^2), \qquad 0 \le z \le 1.$$

The expected area of the new circle is therefore

$$\begin{aligned} E(\pi Z^2) &= 2n(n-1)\pi \int_0^1 \left(z^{2n-1} - z^{2n+1}\right)\mathrm{d}z \\ &= 2n(n-1)\pi \left(\frac{1}{2n} - \frac{1}{2n+2}\right) \\ &= 2n(n-1)\pi \cdot \frac{2}{2n(2n+2)} = \left(\frac{n-1}{n+1}\right)\pi, \end{aligned}$$

which is the required result.

Checks

- With $n = 1$ the first answer is $\pi/2$: a single shot has R^2 uniform on $[0,1]$ (since $P(R^2 \leq u) = u$), so the expected area πR^2 is $\pi/2$.
- With $n = 2$ the second answer is $\pi/3$, the expected area for the *nearer* of two shots; together with the first answer, $2\pi/3$, the two average to $\pi/2$, the single-shot value, as they must.
- A simulation with 2×10^5 trials and $n = 5$ gave expected areas 2.618 and 2.095, against $5\pi/6 \approx 2.618$ and $4\pi/6 \approx 2.094$.

Common slips

- Writing $P(Z \leq z) = z^{2(n-1)}$, as if the remaining shots were a fresh sample of $n-1$; the rejected shot is not random, so the survivors are biased towards O.
- Forgetting the factor n for which shot is the one outside the circle.
- Finding $E(Y)$ and then squaring it: the expected area is $\pi E(Y^2)$, not $\pi\big(E(Y)\big)^2$.

Beyond STEP

These are *order statistics*. Because $U_i = R_i^2$ is uniform on $[0,1]$, the areas are just the order statistics of n uniforms, and the kth smallest of n independent uniforms has a beta distribution with mean $k/(n+1)$; both answers here are the cases $k = n$ and $k = n-1$. The neat picture is that n random points cut $[0,1]$ into $n+1$ gaps of equal expected length. Grimmett and Welsh, *Probability: An Introduction*, covers transformations and order statistics of this kind.

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